Reductant delivery notes may also be accepted for the purposes of verifying that the system has been operated by using reductant. In such cases, the reductant delivery notes should be made available at annual, intermediate

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Reductant delivery notes may also be accepted for the purposes of verifying that the system has been operated by using reductant. In such cases, the reductant delivery notes should be made available at annual, intermediate and renewal surveys.
Where it is proposed to produce aqueous reductant on board, the recording system should consider records of feedstock deliveries and quality;
.12 any other parameter(s) as specified by the applicant; and
RESOLUTION MEPC.399(83) (adopted on 11 April 2025)
2025 GUIDELINES ON SELECTIVE CATALYTIC REDUCTION (SCR) SYSTEMSMEPC 83/17/Add.1
Annex 3, page 10
I:\MEPC\MEPC 83-17-Add.1.docx
.13 a description of a method of storing records should be included for the purpose of maintenance, surveys and inspections:
.1 if paragraph 3.2.8.1 is applied as a means for monitoring catalyst condition/degradation, the readings from the NO x measurement device documenting the deterioration rate of SCR performance, including threshold values, alarms or failure codes; or
.2 if paragraph 3.2.8.2 is applied as a means for monitoring catalyst condition/degradation, records and results of spot checks should be available on board; and

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including threshold values, alarms or failure codes; or
.2 if paragraph 3.2.8.2 is applied as a means for monitoring catalyst condition/degradation, records and results of spot checks should be available on board; and
.3 for the parameter check method as described in paragraph 3.2.8.11, records of reductant composition and quality may be based on delivery notes where these delivery notes include reductant concentration and quality parameters. Reductant delivery notes may also be accepted for the purposes of verifying that the sy stem has been operated using reductant. Where it is proposed to produce aqueous reductant on board, the recording system should consider records of feedstock deliveries and quality.
3.3 Measures to minimize reductant slip
When SCR uses urea solution, ammonia solution or ammonia gas as reductant, measures to prevent reductant slip should be provided to avoid the supply of an excessive amount of reductant in the system. The reductant injection system should be designed to prevent emissions of any harmful substance from the system.
3.4 Pre-certification procedure

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reductant in the system. The reductant injection system should be designed to prevent

emissions of any harmful substance from the system.
3.4 Pre-certification procedure
Test and pre-certification of an engine system fitted with SCR should be conducted either by Scheme A (as given in section 5 of these Guidelines), or by Scheme B (as given in sections 6 and 7 of these Guidelines), as appropriate.
3.5 EIAPP certificate
3.5.1 An Engine International Air Pollution Prevention (EIAPP) certificate (see appendix I of NTC 2008) should be issued by the Administration after approval of the Technical File.
3.5.2 When an applicant chooses Scheme B for pre -certification, the IAPP initial survey should not be completed until the onboard initial confirmation test provides compliant results. The applicant remains the responsible entity until final acceptance of the system.
3.5.3 When the engine is to be certified to both Tier II and Tier III, the EIAPP certificate should be completed for both Tier II and Tier III with a single Technical File covering both Tier modes.
3.5.4 In the context of the EIAPP certificate the term "Engine manufacturer" is the applicant

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should be completed for both Tier II and Tier III with a single Technical File covering both Tier modes.
3.5.4 In the context of the EIAPP certificate the term "Engine manufacturer" is the applicant
for the certification of a system consisting of a marine diesel engine, an SCR chamber and a reductant injection system in accordance with the provisions of paragraph 4.4.4 of NTC 2008.

4 FAMILY AND GROUP CONCEPTS FOR ENGINE SYSTEMS FITTED WITH SCR
4.1 The requirements in chapter 4 of NTC 2008 apply equally to engine systems fitted with SCR.
4.2 The parent engine is to be the engine system fitted with SCR with the highest NO X emission value of the group/family as specified in paragraphs 4.3.9.1 and 4.4.8.1 of NTC 2008. In cases where there is more than one combined engine/SCR system with the same highest NOx emission value given to two decimal places (cycle value in g/kWh) within an engine family or an engine group, the parent engine is the system with the highest raw NO x value emitted

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NOx emission value given to two decimal places (cycle value in g/kWh) within an engine family or an engine group, the parent engine is the system with the highest raw NO x value emitted
from the engine.
4.3 The parent engine for Tier II compliance is not necessarily the same parent of the combined engine/SCR system for Tier III compliance.
5 TEST PROCEDURES FOR SCHEME A
5.1 General
5.1.1 A test for a combined system of an engine fitted with an SCR in Scheme A is to ensure compliance with the applicable NO x emission limits of MARPOL Annex VI, as required. The test bed measurement procedures of chapter 5 of NTC 2008 should apply.
5.1.2 Notwithstanding paragraph 5.1.1, the applicant may choose to test the combined system of an engine fitted with an SCR with a by pass arrangement without that by pass installed for the purpose of test bed measurement. Any effect on the fluid dynamics or reductant distribution caused by the absence of the by -pass arrangement is to be presented

by the applicant.
5.2 Calculation of gaseous emissions
5.2.1 The calculation method in section 5.12 of NTC 2008 is also applied to engine systems

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reductant distribution caused by the absence of the by -pass arrangement is to be presented
by the applicant.
5.2 Calculation of gaseous emissions
5.2.1 The calculation method in section 5.12 of NTC 2008 is also applied to engine systems
fitted with SCR. No allowance is made for the reductant solution injected into the exhaust gas
stream in respect of its effect on exhaust gas mass flow rate calculation (appendix VI) or
dry/wet correction factor (equation (11), paragraph 5.12.3.2.2 of NTC 2008). The NO x
correction factor for humidity and temperature (equations (16) or (17), paragraphs 5.12.4.5
and 5.12.4.6, respectively, of NTC 2008) should not be applied.
5.2.2 For an engine system fitted with SCR, the following parameters should be measured
and recorded in the engine test report in accordance with section 5.10 of NTC 2008:
.1 injection rate of reductant at each load point (kg/h);
.2 exhaust gas temperature at the inlet and outlet of the SCR chamber (°C);
.3 pressure loss (kPa): it is necessary to measure the pressure at the inlet and
at the outlet of the SCR chamber and to calculate pressure loss p. It would

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.2 exhaust gas temperature at the inlet and outlet of the SCR chamber (°C);
.3 pressure loss (kPa): it is necessary to measure the pressure at the inlet and
at the outlet of the SCR chamber and to calculate pressure loss p. It would
also be permissible to measure the pressure loss p of the SCR chamber
with a differential pressure sensor. The allowable p limit should be
confirmed; and
.4 other parameter(s) as specified by the Administration.
RESOLUTION MEPC.399(83) (adopted on 11 April 2025)
2025 GUIDELINES ON SELECTIVE CATALYTIC REDUCTION (SCR) SYSTEMSMEPC 83/17/Add.1
Annex 3, page 12
I:\MEPC\MEPC 83-17-Add.1.docx
6 TEST PROCEDURES FOR SCHEME B
6.1 General
6.1.1 A test for an engine system fitted with SCR in Scheme B is to ensure that the system
complies with the applicable NOx emission limits in MARPOL Annex VI, as required. The test
procedures in Scheme B are as follows:
.1 an engine is tested to obtain the NOx emission value (g/kWh) in accordance
with paragraph 6.2.1 of these Guidelines;
.2 the SCR NOx reduction rate may be calculated by modelling tools, taking into
account geometrical reference conditions, chemical NOx conversion models

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with paragraph 6.2.1 of these Guidelines;
.2 the SCR NOx reduction rate may be calculated by modelling tools, taking into
account geometrical reference conditions, chemical NOx conversion models
as well as other parameters to be considered;

.3 for every type of catalytic element, an SCR chamber, not necessarily to full scale, is to be tested in accordance with section 6.3 of these Guidelines in order to generate data for the calculation model as that used in paragraph 6.1.1.2 of these Guidelines;
.4 the NOx emission from the engine system fitted with SCR, which is calculated in accordance with section 6.4 of these Guidelines using the NO x emission value from the engine and the NO x reduction rate of the SCR chamber ; at this point the Technical File will be completed and this NO x emission value will be entered into the supplement of the EIAPP certificate; and
.5 the NO x emission performance of the engine combined with the SCR is verified by a confirmation test in accordance with the procedure in paragraph 7.5 of these Guidelines.
6.1.2 The calculation of gaseous emissions in paragraph 6.1.1.1 of these Guidelines should

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verified by a confirmation test in accordance with the procedure in paragraph 7.5 of these Guidelines.
6.1.2 The calculation of gaseous emissions in paragraph 6.1.1.1 of these Guidelines should be undertaken in accordance with paragraph 5.2.1 of these Guidelines.
6.2 Verification test procedures for an engine
6.2.1 The purpose of the test of an engine is to establish the emission values for use in